

CS F342 Comp Arch – Midsem Cheatsheet (2021–2025 papers)

Op	Question → Item #
21	11→35, 12→26, 13→3, 14→14, 15→29, 28→19, 26→(hazard: not in syllabus), 22→18, 3→31
22	12→11, 3→31
23	11→35, 12→26, 13→3, 14→14, 15→29, 28→19, 26→(hazard: not in syllabus), 22→18, 3→31
24	11→35, 12→26, 13→3, 14→14, 15→29, 28→19, 26→(hazard: not in syllabus), 22→18, 3→31
25	11→35, 12→26, 13→3, 14→14, 15→29, 28→19, 26→(hazard: not in syllabus), 22→18, 3→31
26	11→35, 12→26, 13→3, 14→14, 15→29, 28→19, 26→(hazard: not in syllabus), 22→18, 3→31
27	11→35, 12→26, 13→3, 14→14, 15→29, 28→19, 26→(hazard: not in syllabus), 22→18, 3→31
28	11→35, 12→26, 13→3, 14→14, 15→29, 28→19, 26→(hazard: not in syllabus), 22→18, 3→31
29	11→35, 12→26, 13→3, 14→14, 15→29, 28→19, 26→(hazard: not in syllabus), 22→18, 3→31
30	11→35, 12→26, 13→3, 14→14, 15→29, 28→19, 26→(hazard: not in syllabus), 22→18, 3→31

log after each heading = number of times it appeared (yr-Q). Same Q again in exam = write the same answer. Occurrences: 1000000, 10 100011, 5w 10101, 4c 000100, add 010000, 1000010, 1000011. Multiple cycles: 1w, 5w, 10, 4, R, add 4, add 6, 3, 3.

A. THEORY

#1. Why opcode + funct fields instead of only opcode? 22-1a

Official key: with only opcode the decoder complexity would increase from 2^{22} to 12^{22} .

A 6-bit opcode can encode only $2^6=64$ instructions. All R-type instrs share opcode 000000 and are told apart by the 6-bit funct → 64 more ALU ops without widening the 32-bit opcode.

R-type needs no 16-bit immediate, so its 6-bit bits are free for shamt/funct; I-types need those bits for imm/address, so they get unique opcodes. Keeps all instrs 32-bit with fields at fixed positions (regularity → simple hardware).

Decoding becomes two-level: main decoder looks only at opcode (6 inputs); ALU decoder uses funct+ALUop → smaller, faster control (see #10).

#2. Why is single-cycle control called a "Truth Table"? 21-1b, 22-1b

Official key: "No next state." In single MIPS every instr finishes in one clock, so the control unit is purely combinational: no state/memory, its outputs (RegWrite, RegDst, ALUSrc, Branch, MemWrite, MemRead, ALUOp, Jump) depend only on the present inputs (opcode, funct, A combinational circuit is completely specified by listing output for every input combination = a truth table (one row per opcode), which is then implemented as SOP logic/PLA/RM. (Multiple cycle control is an FSM → state table, not truth table).

#3. How does the processor capture the side effects when instrs complete? 21-1c

Side effects = changes to architectural state: register file, data memory and PC. Combinational logic computes the new values using the current cycle's data; they are captured only at the rising clock edge by state elements whose write enable is asserted. RF: written if RegWrite=1 (WD3→reg A3), memory written if MemWrite=1 (WD→M[A]), PC←PC every event (multicycle: PCen). Intermediate results in multiple/pipeline are held in non-architectural registers (IR, DR, B, ALUOut / pipeline regs) and committed in the write-back stage.

#4. <2> behaves as <1> – which instrs affected? 21-1d, 22-1d, 24-1d, 25-3b

Shift-by-2 converts a word offset to a byte offset; used only for PC-relative/jump targets. Always state the EFFECT (graded 20% marks for only key: "last 2 bits must be 00").

Official key: BNE, J – instruction alignment problem. beq/one (taken): PCBranch = PC+4 + (SignImm<1>) PC+4+2 imm instead of 4 imm → wrong, half-word aligned address (e.g., imm=3 → PC+4+2 not PC+4+12) → fetches garbage/misaligned instr. Not-taken branch still OK (PC+4).

(1 and 2nd half): JTA = [PC+4:31:28], add<rr> = 26<rr>+1 → 27 bits only; target = half the intended address, wrong/misaligned → jumps to wrong location. Not affected: R-type (add, sub, and, sll, lw, sw, addi – they don't use the shifters (lw/sw offsets are already byte offsets)).

#5. Stuck control signals (single-cycle) 21-1a, 23-1e

Signal	Malfunction & why
MemtoReg	R-type (add, sub, and, or, slt, sll, srl, shll, shllv, shllv2, shllv3, shllv4, shllv5, shllv6, shllv7, shllv8, shllv9, shllv10, shllv11, shllv12, shllv13, shllv14, shllv15, shllv16, shllv17, shllv18, shllv19, shllv20, shllv21, shllv22, shllv23, shllv24, shllv25, shllv26, shllv27, shllv28, shllv29, shllv30, shllv31, shllv32, shllv33, shllv34, shllv35, shllv36, shllv37, shllv38, shllv39, shllv40, shllv41, shllv42, shllv43, shllv44, shllv45, shllv46, shllv47, shllv48, shllv49, shllv50, shllv51, shllv52, shllv53, shllv54, shllv55, shllv56, shllv57, shllv58, shllv59, shllv60, shllv61, shllv62, shllv63, shllv64, shllv65, shllv66, shllv67, shllv68, shllv69, shllv70, shllv71, shllv72, shllv73, shllv74, shllv75, shllv76, shllv77, shllv78, shllv79, shllv80, shllv81, shllv82, shllv83, shllv84, shllv85, shllv86, shllv87, shllv88, shllv89, shllv90, shllv91, shllv92, shllv93, shllv94, shllv95, shllv96, shllv97, shllv98, shllv99, shllv100, shllv101, shllv102, shllv103, shllv104, shllv105, 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